

TCL (Transaction Control Language) and DCL (Data Control Language) Complete Notes

WHAT IS TCL?

TCL stands for Transaction Control Language.

TCL is used to manage transactions in database.

A transaction means:

A group of SQL queries executed together as one unit.

Why TCL is Important?

Suppose:

- Money deducted from one account
- Money added to another account

Both operations should happen together.

If one query fails:

- Database should return to previous state.

This is done using TCL.

Types of TCL Commands

1. BEGIN TRANSACTION / START TRANSACTION
2. COMMIT
3. ROLLBACK
4. SAVEPOINT / SAVE TRANSACTION

1. BEGIN TRANSACTION

Definition

Used to start transaction.

SQL Server Syntax

BEGIN TRANSACTION;

MySQL Syntax

START TRANSACTION;

Purpose

It tells database:

“All next queries belong to one transaction.”

2. COMMIT

Definition

Used to permanently save changes.

Example

BEGIN TRANSACTION;

```
UPDATE customer_orders  
SET order_status = 'Delivered'  
WHERE order_id = 101;
```

COMMIT;

What Happens?

Changes permanently saved in database.

After COMMIT:

- Cannot rollback changes.

3. ROLLBACK

Definition

Used to undo transaction changes.

Example

```
BEGIN TRANSACTION;
```

```
UPDATE customer_orders  
SET total_amount = total_amount - 1000  
WHERE order_id = 101;
```

```
ROLLBACK;
```

What Happens?

Database returns to previous state.

Changes are cancelled.

4. SAVEPOINT / SAVE TRANSACTION

Definition

Used to create checkpoint inside transaction.

Why SAVEPOINT is Used?

Suppose transaction has multiple steps:

- Some steps successful
- One step failed

Instead of cancelling whole transaction,
rollback only some part.

Example (SQL Server)

```
BEGIN TRANSACTION;
```

```
UPDATE inventory
```

```
SET stock = stock - 2  
WHERE product_id = 101;
```

```
SAVE TRANSACTION stock_updated;
```

```
UPDATE customer_orders  
SET order_status = 'Shipped'  
WHERE order_id = 500;
```

```
ROLLBACK TRANSACTION stock_updated;
```

```
COMMIT;
```

What Happens?

- Stock update remains
- Order status update cancelled

Difference Between ROLLBACK and SAVEPOINT

ROLLBACK	SAVEPOINT
Undo full transaction	Undo partial transaction
Returns to beginning	Returns to checkpoint

TCL FLOW

```
BEGIN TRANSACTION  
↓  
INSERT / UPDATE / DELETE  
↓  
SAVEPOINT (optional)  
↓  
COMMIT or ROLLBACK
```

Real-Time Uses of TCL

1. Banking Systems
2. ATM Transactions

3. Online Shopping
4. Ticket Booking
5. Inventory Systems
6. Hospital Billing Systems

Important TCL Points

1. COMMIT saves permanently
 2. ROLLBACK cancels changes
 3. SAVEPOINT creates checkpoints
 4. TCL controls transactions
 5. Transaction ensures data consistency
-

WHAT IS DCL?

Definition

DCL stands for Data Control Language.

DCL is used to control:

- Database permissions
- Security
- User access

Why DCL is Important?

Not every user should:

- Delete records
- Modify salaries
- Access confidential data

DCL controls permissions.

Types of DCL Commands

1. GRANT
2. REVOKE

1. GRANT

Definition

Used to give permissions to users.

Syntax

```
GRANT permission_name  
ON table_name  
TO user_name;
```

Example

```
GRANT SELECT, INSERT  
ON employees  
TO hr_user;
```

What Happens?

User can:

- Read data
- Insert data

User cannot:

- Delete data
- Modify restricted data

2. REVOKE

Definition

Used to remove permissions from users.

Syntax

```
REVOKE permission_name  
ON table_name  
FROM user_name;
```

Example

```
REVOKE INSERT
```

```
ON employees  
FROM hr_user;
```

What Happens?

User can no longer insert records.

Common Permissions in DCL

Permission	Meaning
SELECT	Read data
INSERT	Add records
UPDATE	Modify records
DELETE	Remove records
ALL PRIVILEGES	Full access

Create User Example (MySQL)

```
CREATE USER 'developer'@'localhost'  
IDENTIFIED BY 'dev123';
```

Grant Full Access

```
GRANT ALL PRIVILEGES  
ON company_db.*  
TO 'developer'@'localhost';
```

Remove Full Access

```
REVOKE ALL PRIVILEGES  
ON company_db.*  
FROM 'developer'@'localhost';
```

Real-Time Uses of DCL

1. Company Databases
2. Banking Systems
3. Hospital Management Systems
4. School Management Systems
5. Employee Management Systems

Difference Between TCL and DCL

TCL	DCL
Controls transactions	Controls permissions
COMMIT, ROLLBACK	GRANT, REVOKE
Used for save/undo	Used for security
Maintains consistency	Maintains access control

Interview Questions

TCL

1. What is transaction?
2. Difference between COMMIT and ROLLBACK?
3. What is SAVEPOINT?
4. Why TCL is important?

DCL

1. What is GRANT?
2. Difference between GRANT and REVOKE?
3. What are privileges?
4. Why DCL is used?

One-Line Definitions

TCL

Used to manage transactions in database.

DCL

Used to manage permissions and security in database.